Process Characterization Plan

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCP-005

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2025-11-03	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
ALCOA+	attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CQA	critical quality attribute
CV	column volume
DNA	deoxyribonucleic acid
DS	drug substance
ELISA	enzyme-linked immunosorbent assay
Fab	antigen-binding fragment
Fc	crystallizable fragment
HCP	host cell protein
HMW	high molecular weight species
IAC	intermediate acceptance criterion
LMW	low molecular weight species
MSAT	Manufacturing Science and Technology
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
RSM	response surface methodology
SEC	size exclusion chromatography
SOP	standard operating procedure

1 Purpose and scope

This plan is the protocol for the process characterization of the Protein A chromatography step (Step 5) of the A-Mab drug substance process. It applies the Process Characterization Master Plan PCMP-001 to this step. It adds what is specific to the step, such as the scale-down model, the factors and their ranges, the sampling plan and the values of the acceptance criteria. Where this plan says nothing of its own, PCMP-001 applies.

Protein A chromatography is the capture step of the A-Mab purification train. The resin binds the antibody from the clarified harvest, and most of the host cell protein and DNA leave the column in the flow-through and the wash. The antibody is then eluted at low pH and passed to low-pH viral inactivation (Step 6) without neutralization. The step adds one impurity of its own, which is Protein A ligand that leaches from the resin.

The characterization covers all 6 process parameters of the step. Of these, 4 are studied in a screening design followed by a response-surface design, and 2 are studied one factor at a time. All studies are run in a scale-down model of the commercial column. For pool host cell protein and leached Protein A, which the qualification of the model compares at both scales, each result is translated to manufacturing scale before it is judged against a criterion (Section 5.1). The commercial process is run at the 15,000 L scale at the receiving site in Grafton, WI, which is the scale this characterization supports under the Process Transfer Plan PTP-001.

PCR-005, the Process Characterization Report for this step, will report the results. It will give the proven acceptable ranges (PARs) of the multivariate parameters for each attribute of the Protein A pool with a criterion, and the data for the classification of every parameter of the step. Although Protein A chromatography removes some virus, no viral clearance claim is made for this step (CMC Biotech Working Group 2009). The viral clearance claim of the process rests on low-pH inactivation, anion exchange and virus filtration (PCMP-001, viral clearance claims), and no spiking study is planned here. The life cycle of the resin (its re-use limit, cleaning and sanitization) is governed by the resin life-cycle procedure SOP-2008 and is outside the scope of this plan.

1.1 Changes in this revision

Version 1.0 of this plan was approved on 2025-11-03, before the first study of the step began on 2025-11-05. The studies that this plan governs were executed between 2025-11-05 and 2026-04-17. This Version 2.0 is a protocol amendment issued after

that execution, under the investigation INV-001. QA opened the investigation on 2026-08-03 after its review of PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major.

The investigation found that the criteria used to judge each step in the PAR analysis were not in any approved protocol and had been set after the data were seen (INV-001 finding 1). In this plan, a Version 1.0 analysis limit is a limit that a Version 1.0 report applied when its data were analysed, although no approved protocol stated it. An approved plan criterion is a criterion that an approved version of this plan stated before the studies were run. A Version 2.0 criterion is the drug substance criterion carried back through the platform worst case of each downstream step, and then checked against the A-Mab results of those steps, as PCMP-001 sets out for intermediate acceptance criteria.

For this step, Version 1.0 of this plan contained no approved plan criterion. Version 1.0 of PCR-005 applied one analysis limit of its own, 10,724 ng/mg for pool host cell protein. That limit was the drug substance criterion carried back through the nominal downstream clearance of this campaign and then divided by a safety factor of 4.5. The factor was chosen, not calculated from how the downstream steps perform (INV-001 finding 2). Version 1.0 of PCR-005 judged leached Protein A against the drug substance criterion, and it did not report fragments. Table 1.1 sets the Version 1.0 limits beside the Version 2.0 criteria.

Table 1.1: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, Protein A.

Attribute	Version 1.0[a]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Leached Protein A (ppm)	none[b]	–	≤ 158	Carried back
Fragments, LMW (%)	not assessed[c]	–	≤ 2.45	Carried back
Host cell protein (ng/mg)	≤ 10,724	R1, margin 4.5	≤ 17,069	Carried back, widened[d] from ≤ 16,000

[a] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b] Version 1.0 set no step criterion and judged the step against the drug-substance criterion.
[c] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[d] Widened on verification: a downstream step did better than the platform worst case at the one-sided 95 % prediction bound of a single batch, on an adequate model.
[R1] Version 1.0 rule: nominal back-calculation / margin.

This amendment makes the following changes.

1. It states an intermediate acceptance criterion (IAC) for host cell protein, leached Protein A and fragments in the Protein A pool (Section 8.1). Each is the drug substance criterion carried back through the platform worst case of the downstream steps. Because this derivation uses no data from the Protein A studies, it may be applied to studies that were already executed. The criterion for pool

host cell protein is in force at a wider value, because the A-Mab results of anion exchange showed at the one-sided 95 % prediction bound of a single batch, from a model that passed the adequacy tests, that the step clears more host cell protein than its platform worst case.

2. It qualifies the scale-down model by its offset from manufacturing scale, by the rule that PCMP-001 sets for scale-down model qualification. For each attribute that the qualification compares at both scales and that has a criterion, the offset is subtracted before a result is judged. The ranges and conclusions of PCR-005 for these attributes are therefore stated at manufacturing scale (Section 5.1).
3. It adds fragments measured by nrCE-SDS (AMV-3020, under the non-reduced CE-SDS procedure SOP-3015) as a response of the response-surface design. The method decision AMD-001 of 2026-03-20 set nrCE-SDS as the method for fragments at every step. The retained response-surface samples are tested by nrCE-SDS in study S-PA-RET, the retest of retained samples. The screening samples were not retained. Their SEC fragment results are kept in the record and are not used for any decision, because SEC overstates fragments in the Protein A pool (INV-001 finding 6, Section 4.1).
4. It states the model adequacy criteria and the value that the PAR analysis compares with each criterion, which Version 1.0 left to the analysis (Section 7).
5. It records the execution window of every study that the plan governs (Table 11.1). INV-001 found that the Version 1.0 records gave no execution dates (finding 5).

The Version 2.0 criterion for pool host cell protein, 17,069 ng/mg, is wider than the Version 1.0 analysis limit of 10,724 ng/mg. The Version 1.0 limit was never an approved criterion, so the wider Version 2.0 criterion does not loosen one. From this revision on, a criterion of this plan changes only by the rule for verifying the platform worst cases, which Section 8.1 states.

The factors, their ranges and the designs are those of Version 1.0. No run is added, removed or repeated by this amendment.

2 Objectives

The objectives of the characterization are the following.

1. To estimate, before the first characterization run, the offset between the scale-down model and manufacturing scale for each quantity that the qualification compares, and to qualify the model by that offset for pool host cell protein and leached Protein A (Section 5.1).
2. To identify which of the 4 multivariate parameters, and which of their two-factor interactions, affect pool host cell protein, step yield and leached Protein A (screening design).
3. To fit response-surface models for pool host cell protein, step yield, leached Protein A and fragments by nrCE-SDS over the ranges studied, and to judge their adequacy for prediction.
4. To determine from the response-surface models the PAR of each multivariate parameter for each attribute of the Protein A pool with a criterion, at manufacturing scale where an offset applies.
5. To establish the effect of operating temperature and bed height over their ranges, one factor at a time.
6. To provide the data for the classification of every parameter of the step under the parameter classification procedure SOP-4001.

3 Related documents

Table 3.1 lists the documents of the package that this plan depends on or feeds.

Table 3.1: Documents related to this plan.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-005	Process Characterization Report (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

Table 3.2 lists the controlled procedures and method validations that the studies use. The SOP and AMV numbers are placeholders.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2007	Operation of the Protein A Capture Chromatography Step
	SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
	SOP-3015	Purity by Non-Reduced CE-SDS
Analytical method validation	AMV-3021	Product Titre (Protein A Affinity HPLC)
	AMV-3016	Leached Protein A by ELISA (ppm)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3014	Residual DNA (qPCR)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3020	Purity (non-reduced CE-SDS)
	AMV-3019	Protein Concentration by A280 (UV)

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Protein A capture is an affinity step operated in bind and elute mode. Staphylococcal Protein A, immobilized on the resin, binds the Fc region of the antibody at neutral pH. The contact sits at the interface of the second and third constant domains of the heavy chain. Host cell protein, DNA and most other components of the clarified harvest do not bind and are removed in the load flow-through and the wash. When the pH is lowered, histidine residues at the Fc–ligand interface become protonated, the interaction weakens, and the antibody is released into a small volume of low-pH eluate.

The same capture step has been used to purify the platform antibodies X-Mab, Y-Mab and Z-Mab, and A-Mab through clinical manufacture (CMC Biotech Working Group 2009). Five mechanisms set the expectations for this study.

1. Host cell protein reaches the eluate by two routes. Some binds to the resin backbone or to the ligand. Some associates with the antibody through electrostatic and hydrophobic interactions, which the wash does not disrupt while the antibody stays bound to the ligand. A lower elution pH disrupts more of both kinds of interaction and releases more host cell protein with the product, so pool host cell protein is expected to rise as elution buffer pH falls. In the A-Mab case study pool host cell protein rose with protein load and as elution pH fell (CMC Biotech Working Group 2009), and the same directions are expected here. Pool host cell protein is reported per mass of antibody, and host cell protein that travels with the antibody rises in proportion to it, so that route alone does not explain a rise with load. No mechanism for the rise with load is assumed in this plan. Both parameters act on the host cell protein that is eluted, so an interaction between them is possible, and the designs can estimate it.
2. Step yield is lost when antibody breaks through during the load or remains on the column at the end of collection. Breakthrough rises as the load approaches the dynamic binding capacity and as the residence time falls, because binding is limited by diffusion into the pores of the bead. A higher load flow rate shortens the residence time and lowers the dynamic binding capacity. At high load the bed is then close to its capacity, breakthrough starts earlier and yield falls. At low load the bed has capacity to spare and flow rate has little effect. Yield is therefore expected to fall with flow rate mainly at high load. Extending the end of pool collect recovers the tail of the elution peak and raises yield.
3. Leached Protein A comes from hydrolysis of the coupling chemistry and from proteolysis of exposed ligand. The ligand fragments still bind Fc and elute with the antibody. The amount depends on the coupling chemistry, the cycle number

and the sanitization history of the resin, which are properties of the resin over its life and not of a single run. It also depends on how long the load is held before capture, because proteolytic activity in the clarified harvest is expected to rise while it is held, so that a load held longer cleaves more ligand from the resin. Leached Protein A is reported per mass of antibody (ppm). If a similar amount of ligand leaches in each cycle, leached Protein A in ppm falls as protein load rises, and it is expected to depend little on the other parameters within their ranges.

4. An antibody fragment is captured only if it still carries Fc. Fab fragments flow through with the load, and the pool therefore carries less fragment than the feed. A lower elution pH may add a little acid-catalysed hinge hydrolysis during elution. The pool also still carries host cell protein, and the part of it that elutes in the low molecular weight window of SEC is read as fragment. For this reason SEC overstates fragments in the Protein A pool, and nrCE-SDS does not (INV-001 finding 6).
5. DNA is polyanionic and does not bind Protein A. Its clearance is large and is not expected to depend on the operating parameters within their ranges.

Protein A, when operated under platform conditions, does not separate glycan variants or charge variants, because it binds the Fc region independently of both (CMC Biotech Working Group 2009). Glycan variants and charge variants are therefore not responses of the designs.

Aggregate and monomer both carry Fc and both bind Protein A. Aggregates bind with higher avidity and can be partly resolved on elution, but under platform conditions the aggregate level of the pool follows that of the load. Low pH can partly unfold the antibody and promote aggregation, and the amount formed grows with the time spent at low pH. In the column the antibody is at elution pH only for the duration of the elution. In the Protein A characterization of the case study no process parameter had a significant effect on aggregate (CMC Biotech Working Group 2009). Aggregate formation at low pH is characterized in the low-pH viral inactivation study (PCP-006). That study spans pH 3.2–4.0 for 60–180 min, which covers the elution pH range of this step for a much longer time. For these reasons aggregate is not a response of the designs of this plan. It is measured on every pool by SEC for information.

4.2 Quality attributes in scope

Leached Protein A is the only CQA that this step introduces. It enters the process at capture. The step is also the principal point of clearance for host cell protein and DNA, and it partly clears fragments. Table 4.1 lists the attributes in scope with their drug substance criteria and their criticality from the CQA register of PCMP-001. Criticality was assigned with Tool #1 of the A-Mab case study, whose score is the product of impact and uncertainty (CMC Biotech Working Group 2009). The levels run from very low (VL) to very high (VH), and the Tool #1 column gives the score behind each level.

Table 4.1: Quality attributes in scope of this plan, with the drug substance criterion and criticality.

CQA	Category	DS criterion	Criticality	Tool #1
Leached Protein A	process impurity	≤ 5 ppm	L-M	16
Host Cell Protein (HCP)	process impurity	≤ 100 ng/mg	M-H	36
Fragments (LMW, clipped species)	size variant	≤ 2 % (nrCE-SDS)	L	12
Residual DNA	process impurity	≤ 0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	≤ 5 % HMW (SEC)	H	60

The drug substance criteria do not apply to the Protein A pool directly. Cation exchange and anion exchange remove host cell protein and leached Protein A after this step, so a pool that the later steps bring within the drug substance limit can itself lie far above it. Host cell protein, leached Protein A and fragments in the Protein A pool are therefore each judged against an intermediate acceptance criterion (Section 8.1). Residual DNA is measured for information and is not a response of the designs. DNA is polyanionic and does not bind Protein A, so it leaves the column in the flow-through and the wash, and its clearance is expected to be large and insensitive to the parameters. Its drug substance criterion of ≤ 0.001 ng/dose is taken from the A-Mab case study (Table 2.17), where it is set far below the limit of the World Health Organization for DNA from continuous cell lines (CMC Biotech Working Group 2009). Aggregate is measured by SEC for information only. Step yield is a performance attribute. It has no acceptance criterion and is used for the classification of parameters that act on process performance.

4.3 Risk-based prioritization of parameters

The pre-characterization risk assessment RA-001 assessed every parameter of the step before characterization began. Each parameter was ranked on its potential impact on the CQAs and on process performance, and on its potential to interact with other parameters.

Protein load, elution buffer pH, load flow rate and end of pool collect were assigned to multivariate study. Pool host cell protein is expected to rise with protein load and as elution buffer pH falls, and the two may interact (Section 4.1). Step yield is expected to fall as load flow rate rises and to rise with end of pool collect. Load flow rate is expected to interact with protein load, because a shorter residence time lowers the dynamic binding capacity and costs yield only when the load is close to it. A design that varies the four parameters together is needed to estimate these interactions.

Operating temperature was assigned to univariate study. A higher temperature speeds the diffusion of the antibody into the pores, which raises the dynamic binding capacity, and it strengthens the hydrophobic contacts at the Fc–ligand interface, which makes elution slightly less complete. A lower temperature acts the opposite way on both. Because temperature acts on the dynamic binding capacity, it could interact with protein load in the same way as load flow rate does. Over the range studied, however, its effect on binding capacity is expected to be small against that of residence time, and the resin manufacturer specifies binding capacity across that range. An interaction of temperature with protein load is therefore expected to be small, and on that basis temperature is studied one factor at a time. The univariate study runs at the set-point load, so an interaction of temperature with a high load is not estimated (Section 6.4).

Bed height was also assigned to univariate study, although it too can interact with protein load. At constant linear velocity bed height sets the residence time, and a shorter residence time lowers the dynamic binding capacity (Section 4.1). The designs study the interaction of residence time with protein load through load flow rate, at the set-point bed height. Bed height itself is fixed for each packed column by the packing specification and does not vary from run to run in manufacture, so it is studied one factor at a time. The shortest bed of the range, 10 cm at the set-point velocity, gives a residence time of 3 min, shorter than any run of the designs. A short bed combined with a high protein load, where breakthrough is most likely, is therefore not studied, and no claim is made for it.

5 Materials and methods

5.1 Scale-down model and its qualification

The scale-down model is a laboratory column packed with the commercial Protein A resin. It is designed on the scaling principles that PCMP-001 sets for scale-down models. The bed height and the residence time of the commercial column are kept. Load, wash and elution volumes are normalized to column volumes, and column efficiency is checked by plate count and peak asymmetry after packing. Buffers are made to the commercial recipes. The model is operated at the set-point of every parameter of Table 6.1 unless a design sets it otherwise.

The qualification runs are made under the scale-down model qualification procedure SOP-1001 in study S-PA-SDM, before the first run of the screening design, and they give the offset of each quantity the qualification compares. Whether the model is qualified for an output is decided at analysis, because the decision needs the response-surface model of the step. In that study, 3 runs of the model at set-point are compared with 5 manufacturing-scale batches of A-Mab run at set-point. The quantities compared are pool host cell protein, leached Protein A, step yield and column efficiency (plate count and peak asymmetry).

The model is qualified by the rule that PCMP-001 sets for scale-down model qualification, which judges the model by its offset from manufacturing scale. The scale-down model need not reproduce the manufacturing mean, but the difference between the two scales must be known well enough that no conclusion depends on its exact size. The offset of a quantity is the mean of the scale-down runs minus the mean of the manufacturing-scale batches. It is estimated with its two-sided 90 % confidence interval, each end of which is a one-sided 95 % confidence bound, the level used to compare two means at 5 % on each side. For a quantity that is compared at both scales and has a criterion, each scale-down result is translated to manufacturing scale by subtracting the offset before it is judged. The model is qualified for such a quantity when its conclusion over the NOR is the same with the offset at either end of the confidence interval. That conclusion is whether the Protein A pool meets the criterion at every combination of NOR low, set-point and NOR high of the 4 response-surface factors. A model whose conclusion for a quantity changes between the two ends of the interval is not used for that quantity.

For this step the rule applies as follows.

1. Pool host cell protein and leached Protein A have a criterion at the step (Table 7.1). Their offsets are subtracted before each result is judged, and the model is qualified or not for each of them by the rule above.

2. Step yield and column efficiency have no criterion at the step. Their offsets are recorded in PCR-005, and no conclusion depends on them.
3. Fragments are not measured in the qualification. AMD-001 was taken on 2026-03-20, after the scale-down model qualification S-PA-SDM had closed on 2025-11-14. The qualification therefore has no fragment result at either scale, and fragments carry no offset. They are judged as the response-surface model predicts them.

Leached Protein A is expected to be lower in the laboratory column than at manufacturing scale. The manufacturing load is held longer before capture, so it is expected to cleave more ligand from the resin (Section 4.1, item 3). This difference is the offset of leached Protein A, and it is subtracted as every offset is. Where the cause of an offset is known, PCMP-001 asks the report to state it.

Fragment results from the scale-down model are accepted on mechanism. The capture of Fc-bearing species depends on bed height and residence time, which the model keeps at their commercial values. The low-pH hinge hydrolysis depends on pH, time at low pH and temperature, and the model uses the commercial elution buffer at the same temperature and the same residence time. The plan has no fragment result at manufacturing scale with which to confirm the mechanism directly. The confirmation it relies on is indirect. The qualification compares the plate count and peak asymmetry of the laboratory column with those of the commercial column, which shows whether the laboratory bed behaves as the commercial bed does. PCR-005 tabulates the offsets of this step, and PCMR-001 summarises the offsets of the whole train.

5.2 Operation

Each run follows the Protein A operating procedure SOP-2007 at laboratory scale. The column is equilibrated, loaded with clarified harvest to the target protein load, washed and eluted with the elution buffer at the pH set by the design. The eluate is collected from the rise of the absorbance at 280 nm until the end of pool collect set by the design, expressed in column volumes. The column is then stripped, cleaned and stored as the resin life-cycle procedure SOP-2008 specifies. The pool is not neutralized, because the commercial eluate passes to low-pH viral inactivation at elution pH.

Each design is loaded from one pool of clarified harvest produced at set-point conditions and held frozen in aliquots. A single feed keeps feed variation out of the estimated effects. The effect of feed quality on the step is therefore not estimated in this plan. Runs are executed in random order within each design.

5.3 Analytical methods

Each response is measured by a validated method under its own AMV. Table 5.1 lists the methods, the samples on which each is run and the purpose of each measurement.

Table 5.1: Analytical methods, samples and purpose of each measurement. The method validations are listed in the procedures table of Section 3.

Attribute	Method	Samples	Purpose
Pool host cell protein (ng/mg)	Host-Cell Protein ELISA	Load and pool of every run	Response of both designs
Leached Protein A (ppm)	Leached Protein A by ELISA (ppm)	Pool of every run	Response of both designs
Step yield	Protein Concentration by A280 (UV)	Load and pool of every run	Response of both designs
Fragments, LMW (%)	Purity (non-reduced CE-SDS)	Retained pools of the response-surface design	Response of the response-surface design
	Size-Variants (SEC-HPLC)	Pool of every run of both designs	For information only
Aggregates, HMW (%)	Size-Variants (SEC-HPLC)	Pool of every run	For information only
Residual DNA	Residual DNA (qPCR)	Load and pool of the centre-point runs	For information only

Pool host cell protein is reported per mass of antibody in the pool, so that it does not depend on how far the pool is diluted by the end of pool collect. Step yield is the mass of antibody in the pool divided by the mass loaded, both from the protein concentration by absorbance at 280 nm and the measured volumes.

Fragments are measured by nrCE-SDS, which PCMP-001 sets as the method for fragments at every step. SEC is kept as the method for aggregate, because the co-eluting host cell protein falls in the low molecular weight window and not in the aggregate window. SEC fragment results are recorded for every run of both designs, but they are not used for any decision (Section 4.1). Table 5.2 shows which method measured fragments in each designed study of the step.

Table 5.2: Methods used for size variants in the designed studies, Protein A.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW	SEC	
Fragments, LMW	SEC[a)]	SEC[a)], nrCE-SDS[b)]

[a)] In the Protein A and low-pH viral inactivation pools part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[b)] Retained samples of the response-surface design, tested 2026-04-06 to 2026-04-17.

5.4 Sampling plan

The load of each design is sampled once per feed aliquot thawed. The pool of every run is sampled once at the end of collection, after mixing, and its volume is recorded. Each pool sample is divided into aliquots for each method of Table 5.1. One further aliquot of every pool of the response-surface design is retained frozen under the sample management procedure of Analytical Development, so that a method decided later can be applied to the same samples. The retained pools are the samples tested by nrCE-SDS in study S-PA-RET.

Every sample is labelled with the study, the run number of the design and the sample point, and the chain of custody is recorded from sampling to result.

5.5 Statistical methods

The designs are built and analysed under the DoE procedure SOP-1002 with the common statistical approach of PCMP-001. Factors are coded from -1 to $+1$ over their range studied, with 0 at the centre. For each of the 4 design factors the centre of the range studied is the set-point, so the centre-point runs of both designs are set-point runs.

The screening data are fitted with a model of main effects and two-factor interactions, which has 11 terms. Effects are reported as the change in the response over the coded range of the factor, which is twice the coefficient, and significance is judged at $\alpha = 0.05$. The screening model identifies effects. A two-level design cannot resolve curvature, so the screening model is not used to predict between its corners.

The response-surface data are fitted with a full quadratic model of 15 terms. The response-surface model is the predictive model of the step, and every prediction and every PAR in PCR-005 comes from it. Each fitted model is reported with R^2, adjusted R^2 and predicted R^2, the model F-test, and the test of lack of fit against pure error from the centre points. Residuals are plotted against predicted values, against run order and on a normal probability plot. Where a model fails the adequacy criteria of Section 7, a transformation of the response or a justified change of terms may be applied, and the change is reported with its reason.

A prediction of pool host cell protein or leached Protein A is translated to manufacturing scale by subtracting the offset of that attribute before it is compared with a criterion (Section 5.1). The PAR analysis is described in Section 8. Where no adequate model results for a response, that response is reported descriptively against its criterion over the runs of the design, and no PAR is claimed from it.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 lists the parameters of the step with their set-point, the range to be studied, the NOR and the type of study. The classification of each parameter is a result of the study and is not given here.

Table 6.1: Parameters of the Protein A step to be studied, with set-point, range studied, NOR and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	30	10–50	20–40[a)]	multivariate
Elution buffer pH	pH	3.55	3.2–3.9	3.4–3.7	multivariate
Load flow rate	cm/hr	200	100–300	150–250	multivariate
End of pool collect	CV	2.6	2–3.2	2.3–2.9	multivariate
Operating temperature	°C	22	15–30	18–26	univariate
Bed height	cm	20	10–30	15–25	univariate

[a)] The NOR against which the studies were planned and judged. PCR-005 Version 2.0 narrowed protein load to 20–39 g/L resin.

The ranges studied are between 1.9 and 2.3 times the width of the NOR. A range wider than the NOR shows how the step behaves near and beyond the edges of routine operation, and it gives the space within which a PAR can be claimed. A PAR is never claimed beyond the range studied (PCMP-001, proven acceptable ranges).

Load flow rate is set as a linear velocity, so at the set-point bed height of 20 cm it fixes the residence time. The residence time is 6 min at the set-point of 200 cm/hr. Over the range studied it runs from 4 min at the highest flow to 12 min at the lowest. The upper end of the flow range therefore tests the shortest residence time against the highest load, which is the condition most likely to cause breakthrough.

The lower end of the elution pH range, pH 3.2, tests the strongest elution and the largest release of weakly bound host cell protein. The upper end, pH 3.9, tests the weakest elution, with the broadest and latest peak and the least co-eluting host cell protein.

Table 6.2 gives the letter code of each design factor, which the design matrices in the appendices use. Every design factor is studied in both the screening and the response-surface design.

Table 6.2: Letter codes of the design factors.

Code	Factor	Unit
A	Protein load	g/L resin
B	Elution buffer pH	pH
C	Load flow rate	cm/hr
D	End of pool collect	CV

6.2 Screening design

The screening design (study S-PA-SCR) is a full two-level factorial design in the 4 factors of Table 6.2, with 16 factorial runs and 3 centre points, 19 runs in all. Appendix A gives the planned matrix in coded units. Because the design is a full factorial, no effect is aliased with another. Every main effect and every two-factor interaction can be estimated. An interaction of protein load with elution buffer pH is possible and one with load flow rate is expected (Section 4.1), so both must be estimable.

The interaction model leaves 8 residual degrees of freedom. Of these, 2 come from the replicated centre points as pure error and 6 are available to test lack of fit. The centre points also allow a test for curvature. The responses are pool host cell protein, step yield and leached Protein A. Fragments by SEC are recorded for every run for information.

All 4 factors are carried into the response-surface design whatever the screening result. Each of them carries an effect expected in Section 4.1, and 4 factors are few enough for a central composite design. The screening design therefore serves to identify effects and interactions and to check the ranges before the larger design is run.

6.3 Response-surface design

The response-surface design (study S-PA-RSM) is a face-centred central composite design in the same 4 factors. It has 16 factorial runs, 8 axial runs at the faces of the cube (coded -1 and $+1$) and 4 centre points, 28 runs in all. Appendix B gives the planned matrix. A face-centred design keeps every run inside the range studied, so no axial run exceeds a range that RA-001 and the platform history support.

The full quadratic model leaves 13 residual degrees of freedom, with 3 for pure error and 10 for lack of fit. The responses are pool host cell protein, step yield, leached Protein A and fragments by nrCE-SDS. Fragments are measured on the retained pools in study S-PA-RET (Section 5.4).

6.4 Univariate assessment

Operating temperature and bed height are studied one factor at a time (study S-PA-UNI), with the other parameters at their set-points. Each is run in duplicate at the low end and at the high end of its range studied. The set-point runs of the scale-down model qualification (Section 5.1) serve as the reference. Operating temperature is studied from 15–30 °C around a set-point of 22 °C. The column and all buffers are equilibrated to the run temperature. Bed height is studied from 10–30 cm around a set-point of 20 cm, with the linear velocity held at its set-point, so that the residence time changes with the bed height. Holding the velocity makes the short bed the more severe test, because the short bed then also has the shortest residence time.

The responses are pool host cell protein, step yield and leached Protein A. Fragments are not measured in the univariate study, which ran before AMD-001. A univariate study has no model, so there is no prediction interval to compute, and each individual result stands for a single batch. The range of a univariate parameter is therefore supported when every individual result at both ends meets the criterion of Table 7.1 for pool host cell protein and for leached Protein A, after the offset is subtracted. Step yield has no criterion and is reported as the size of the effect only. A univariate study cannot detect an interaction. The ranges of operating temperature and bed height are therefore supported only with the other parameters at their set-points, and PCR-005 states this beside each result.

7 Acceptance and decision criteria

These criteria decide when the scale-down model is qualified, when a model may be used for prediction, when a range is proven acceptable and how a parameter is classified. Criteria 2 to 5 are the model adequacy criteria of PCMP-001. Their limits are the campaign thresholds that PCMP-001 sets for every step alike. These are the conventional 5 % significance level for the tests, and a limit of 0.20 on how far predicted R^2 may fall short of R^2, which guards against a model with more terms than the data support.

1. Scale-down model qualification. For pool host cell protein and for leached Protein A, the model is qualified when the conclusion of criterion 6, whether the Protein A pool meets its criterion across the NOR, is the same with the offset at either end of its 90 % confidence interval. A model that is not qualified for an attribute is not used for that attribute. Step yield and column efficiency have no criterion at the step, so their offsets are recorded only. Fragments carry no offset, because the qualification has no fragment result, and they are accepted on the mechanism stated in Section 5.1.
2. Model significance. A response-surface model is used for prediction only if the model F-test is significant at $\alpha = 0.05$.
3. Lack of fit. The test of lack of fit against pure error is not significant at $\alpha = 0.05$.
4. Predictive power. The predicted R^2, computed from the prediction error sum of squares (PRESS), falls short of the R^2 of the same model by no more than 0.20.
5. Residuals. A regression of the absolute residuals on the predicted values, and a regression of the residuals on run order, each show no slope significant at $\alpha = 0.05$. No run has a Cook’s distance above 4/n, a common screening cut-off for influential runs that scales with the number of runs, which is 0.14 for the 28 runs of the response-surface design.
6. Protein A pool and PARs. Each attribute of the Protein A pool with a criterion is judged against the criterion of Table 7.1 by the analysis of Section 8, after the offset is subtracted where one applies. A range is proven acceptable for an attribute when the NOR-propagated analysis (a simulation in which the other factors vary within their NOR, Section 8.2) meets the criterion across it.
7. Classification. Each parameter is classified under the parameter classification procedure SOP-4001 by the decision logic of PCMP-001. A parameter whose effect on a CQA is significant and practically relevant is critical, and it is well controlled when it has a low risk of leaving its acceptable range in operation. A parameter that affects only process performance is a key parameter, and one with no relevant effect is a general parameter. The size of each effect is stated beside its significance. An effect is practically relevant for an attribute when the NOR-propagated PAR of the parameter is narrower than its range studied.

A significant effect whose PAR spans the whole range studied does not by itself make the parameter critical.

A run lost or invalidated during execution is reported in PCR-005, with a statement of whether the design was repaired or re-executed.

Table 7.1: Criteria for the Protein A pool, with their basis, the statistic judged and the scale-down offset.

Attribute	Criterion	Basis	Statistic judged	Offset
Pool HCP (ng/mg)	≤ 17,069	Carried back[a], widened[b]	Single batch[c]	Subtracted
Leached Protein A (ppm)	≤ 158	Carried back[a]		
Fragments (LMW, %, nrCE-SDS)	≤ 2.45	Carried back[a]		None[d]
Step yield	none	Performance attribute	not applicable	Recorded only

[a] The drug substance criterion carried back through the platform worst case of the downstream steps (PCMP-001, intermediate acceptance criteria).
[b] Widened, because the A-Mab results of anion exchange showed at the one-sided 95 % prediction bound of a single batch, from a model that passed the adequacy tests, that the step clears more host cell protein than its platform worst case (PCMP-001, verification of the platform worst cases).
[c] The value compared with the criterion is the upper end of the 95 % prediction interval of a single batch. The criterion is an upper limit, so only that end is compared, which makes it a one-sided 97.5 % prediction limit.
[d] The scale-down model qualification has no fragment result, so fragments carry no offset.

8 Proven acceptable ranges (planned analysis)

8.1 Intermediate acceptance criteria

A drug substance criterion does not say what the Protein A pool must meet, because the steps after capture remove part of what the pool carries. Each attribute of the Protein A pool is therefore judged against an intermediate acceptance criterion (IAC), derived by the rule that PCMP-001 sets for these criteria. The drug substance criterion is carried backward through the train, using the worst case that each downstream step showed on the platform with X-Mab, Y-Mab and Z-Mab. For a step that removes an attribute the worst case is the lowest clearance fold. For fragments, which are a percentage, the clearance fold of a step is the fragment level in its load divided by the fragment level in its pool. For a step that adds to an attribute the worst case is the largest increase. No margin is applied. Table 8.1 gives the three criteria of this step with their derivation.

Table 8.1: Intermediate acceptance criteria for the Protein A output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Host cell protein (ng/mg)	≤ 100	171	0.00	≤ 17,069	Carried back, widened[a)]
Fragments, LMW (%)	≤ 2.00	1.30	0.15	≤ 2.45	Carried back
Leached Protein A (ppm)	≤ 5.00	31.6	0.00	≤ 158	Carried back

[a)] The one-sided 95 % prediction bound of a single batch at a downstream step's A-Mab worst case within its NOR was better than the platform worst case. The bound replaced it.

DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.

Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

For pool host cell protein the drug substance limit of 100 ng/mg is carried back through cation exchange and anion exchange, which together clear host cell protein 170.690-fold in the derivation. That gives 17,069 ng/mg. For leached Protein A the

limit of 5 ppm and a combined clearance of 31.6-fold give 158 ppm. For fragments the limit of 2.00 % is multiplied by the clearance fold of cation exchange, 1.300, and the largest increase of low-pH viral inactivation, 0.15 percentage points, is subtracted. That gives 2.45 %.

The criterion for pool host cell protein is in force at a widened value. From the platform worst cases alone, the derivation gave 16,000 ng/mg. The A-Mab model of anion exchange passed the adequacy tests, and it showed at the one-sided 95 % prediction bound of a single batch that the step clears host cell protein better than its platform worst case. That bound replaced the platform value, which widened the criterion to the value above. The criterion for fragments is at its a priori value of 2.45 %, because the A-Mab results of the downstream steps did not replace any platform value for fragments (Table 8.2). The criterion for leached Protein A is at its a priori value of 158 ppm, because leached Protein A is not a response of the cation exchange or anion exchange designs, so its platform values cannot be checked in this campaign.

PCMP-001 sets, in its rule for verifying the platform worst cases, how the A-Mab results of a downstream step change a criterion of this step. The rule was set before any Version 2.0 report applied it.

1. Where a downstream step does worse within its NOR than its platform worst case, the one-sided 95 % prediction bound of a single batch at its A-Mab worst case, taken on the unfavourable side, replaces the platform value. The criterion of this step then becomes stricter.
2. Where the one-sided 95 % prediction bound of a single batch at the A-Mab worst case within the NOR is better than the platform value, and the model of the downstream step passes the adequacy tests of PCMP-001 (no significant lack of fit, and predicted R^2 within the limit of R^2), the bound replaces the platform value. The criterion of this step then becomes wider. The point prediction is never used to widen a criterion.
3. Otherwise the platform value stands.

PCR-005 relies on the widened criterion for pool host cell protein. For it, PCR-005 gives the following.

1. The bound and the platform value it replaced.
2. The runs of the anion exchange design behind the bound.
3. The adequacy results of the anion exchange model.
4. The reason why the performance of anion exchange for A-Mab is known well enough to replace the platform value.

Table 8.2 lists the downstream values that the criteria of this step rest on, with the platform value, the value in force and the report that checks each.

Table 8.2: Downstream values on which the criteria of this step rely, the value in force and the report that checks each.

Downstream step	Attribute	Worst case	Platform value	Value in force	Value set by	Checked in
Low-pH viral inactivation	Fragments, LMW	Largest increase (percentage points)	0.15	0.15	Platform[a)]	PCR-006
Cation exchange	Host cell protein	Lowest clearance fold	40.0	40.0	Platform[a)]	PCR-007
	Fragments, LMW		1.30	1.30	Platform[a)]	PCR-007
	Leached Protein A		3.16	3.16	Platform[b)]	not checked
Anion exchange	Host cell protein	Lowest clearance fold	4.00	4.27	Bound, widened[c)]	PCR-008
	Leached Protein A		10.0	10.0	Platform[b)]	not checked

[a)] The A-Mab result confirmed the platform worst case, which stands.
[b)] Not a response of the downstream designs, so the platform value cannot be checked in this campaign and stands.
[c)] The one-sided 95 % prediction bound of a single batch at the A-Mab worst case within the NOR was better than the platform value, and the model passed the adequacy tests. The bound replaced the platform value.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

8.2 The two analyses

A PAR is computed for each attribute of Table 7.1 that has a criterion and for each of the 4 factors of the response-surface model. Two analyses are run, as PCMP-001 defines them for proven acceptable ranges. In both, a prediction of pool host cell protein or leached Protein A is translated to manufacturing scale by subtracting its offset before it is compared with the criterion (Section 5.1).

1. The set-point analysis is a one-factor scan of the model's mean prediction. It holds the other factors at their set-points and scans the factor across its range studied. The PAR is the contiguous range around the set-point over which the mean prediction of the response-surface model meets the criterion.
2. The NOR-propagated analysis is a Monte Carlo simulation in which the other factors vary as they would in routine operation. It scans the factor at 81 grid points across its range studied. At each grid point it draws 2,000 operating points for the other factors. Each factor is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR, and clipped to the range studied. The model residual and the uncertainty of the fitted coefficients are included. The PAR is the contiguous range around

the set-point over which the 95 % prediction interval of a single batch meets the criterion.

The NOR-propagated PAR is the one that supports a claim. It is in general narrower than the set-point PAR, because the range must hold while the rest of the operating point moves within its NOR. A response-surface model predicts the mean response, and at the edge of a region defined on mean predictions a single batch meets its criterion only about half the time (CMC Biotech Working Group 2009).

8.3 Decision rule and reporting

A parameter that affects several attributes has a PAR for each, and the range carried into the control strategy is their intersection. Where the set-point itself fails a criterion, PCR-005 reports that no PAR exists for that attribute and states the consequence for the NOR. A PAR that reaches the edge of the range studied is reported as reaching it, and no claim is made beyond it.

PCR-005 reports both analyses for every attribute and factor in one table. For each attribute with a criterion it shows, as a two-panel figure, the PAR of the factor with the largest main effect on that attribute in the response-surface model. The left panel shows the set-point analysis and the right panel the NOR-propagated analysis, with the region that meets the criterion shaded green. The univariate parameters have no model-based PAR. The ranges of operating temperature and bed height are supported by the measured responses against the same criteria at each end of the range studied.

9 Data management and integrity

All data are recorded to ALCOA+ principles. Run records are kept in the electronic laboratory notebook, and analytical results in the laboratory information management system of Analytical Development. The design files and the analysis scripts are held under version control, and every model in PCR-005 is reproducible from the recorded data and the script version.

Each result is second-person verified before it enters the analysis. An excluded result is reported with the reason, and no result is excluded on the basis of its value alone. The execution dates of every study and the calibration status of any instrumented equipment are recorded in the study record. Results obtained before AMD-001, and the SEC fragment results of the screening design, stay in the record with their status marked.

10 Roles and responsibilities

Table 10.1 gives the responsibility of each function for this plan.

Table 10.1: Responsibilities of the functions for this plan.

Function	Responsibility
Process Development / MSAT	Author of this plan and of PCR-005. Executes the runs and the scale-down model qualification.
Analytical Development	Runs the methods of the analytical methods table, the retest of retained samples and sample management.
Manufacturing Science	Supplies the manufacturing-scale data for the offsets of the scale-down model and reviews transfer of the ranges to Grafton.
Statistics	Reviews the designs, the models, the adequacy criteria, the offsets and the PAR analysis.
Quality Assurance	Approves this plan and PCR-005, and closes the actions of INV-001 that concern this step.

11 Deliverables and schedule

The deliverable of this plan is PCR-005, the Process Characterization Report for the Protein A step, with its ground-truth annex. PCR-005 reports the offsets and the qualification of the scale-down model, both designs, the univariate studies and the PARs, and it proposes the classification of every parameter. It rolls up into PCMR-001.

The studies run in this order. The scale-down model is qualified first. The screening design follows, then the univariate study and the response-surface design. The retained pools are tested by nrCE-SDS after AMD-001. Table 11.1 records the execution window of each study, because INV-001 found that the Version 1.0 records gave none (finding 5).

Table 11.1: Execution window of each study of the Protein A step governed by this plan.

Study	Study type	Start	End
S-PA-SDM	scale-down model qualification	2025-11-05	2025-11-14
S-PA-SCR	screening design	2025-11-17	2025-12-05
S-PA-UNI	univariate study	2025-12-08	2025-12-12
S-PA-RSM	response-surface design	2026-01-12	2026-02-06
S-PA-RET	retest of retained samples	2026-04-06	2026-04-17

12 Risks and assumptions

The studies carry the following risks and rest on the following assumptions.

1. The scale-down model represents the commercial column within a known offset. For pool host cell protein and leached Protein A this is tested by the qualification rule of Section 5.1. With 3 scale-down runs and 5 manufacturing batches the confidence interval of an offset may be wide. If the conclusion for one of these attributes then differs between the two ends of its interval, the model is not used for it, and its ranges cannot be set from the designs. For fragments the model rests on mechanism, because capture depends on bed height and residence time, and hinge hydrolysis on pH, time and temperature, all of which the model keeps at commercial values. The qualification has no fragment result at either scale with which to confirm this directly.
2. Leached Protein A is expected to be lower in the laboratory column than at manufacturing scale, because the manufacturing load is held longer before capture (Section 4.1, item 3). This difference is the offset of leached Protein A, and it is subtracted before any result for leached Protein A is judged.
3. Each design uses one feed. The effect of feed host cell protein, titre and viability on the step is not estimated here. The case study reports that Protein A capture is robust over a wide range of feed inputs (CMC Biotech Working Group 2009), and the feed is characterized in PCP-003 and PCP-004.
4. The criteria of this step rest on the downstream values of Table 8.2. If the verification of a downstream step changes one of them before PCR-005 is approved, PCR-005 uses the criterion then in force.
5. Fragments by nrCE-SDS rest on the retained pools of the response-surface design. If a retained pool is lost or found unsuitable, that run has no nrCE-SDS result. The fragment model is then fitted to the remaining runs only if the design still supports it, and otherwise fragments are reported descriptively.
6. Supply of clarified harvest and resin may limit the number of runs. The designs are executed in full before analysis, and no run is dropped to save material.

13 Approvals

This plan is approved by the functions listed in the approval record at the front of this document. Version 2.0 takes effect on 2026-09-30. It replaces Version 1.0 for the analysis and the reporting of the studies of the step.

Appendix A. Planned screening design

Table 13.1 gives the planned screening design (study S-PA-SCR) in coded units. The factor codes are those of Table 6.2. Runs are listed in standard order, grouped by run type, and executed in random order.

Table 13.1: Planned screening design in coded units (standard order).

Run	Type	A	B	C	D
1	factorial	−1	−1	−1	−1
2		−1	−1	−1	+1
3		−1	−1	+1	−1
4		−1	−1	+1	+1
5		−1	+1	−1	−1
6		−1	+1	−1	+1
7		−1	+1	+1	−1
8		−1	+1	+1	+1
9		+1	−1	−1	−1
10		+1	−1	−1	+1
11		+1	−1	+1	−1
12		+1	−1	+1	+1
13		+1	+1	−1	−1
14		+1	+1	−1	+1
15		+1	+1	+1	−1
16		+1	+1	+1	+1
17	centre	0	0	0	0
18		0	0	0	0
19		0	0	0	0

Appendix B. Planned response-surface design

Table 13.2 gives the planned face-centred central composite design (study S-PA-RSM) in coded units. Runs are listed in standard order, grouped by run type, and executed in random order.

Table 13.2: Planned response-surface design in coded units (standard order).

Run	Type	A	B	C	D
1	factorial	−1	−1	−1	−1
2		−1	−1	−1	+1
3		−1	−1	+1	−1
4		−1	−1	+1	+1
5		−1	+1	−1	−1
6		−1	+1	−1	+1
7		−1	+1	+1	−1
8		−1	+1	+1	+1
9		+1	−1	−1	−1
10		+1	−1	−1	+1
11		+1	−1	+1	−1
12		+1	−1	+1	+1
13		+1	+1	−1	−1
14		+1	+1	−1	+1
15		+1	+1	+1	−1
16		+1	+1	+1	+1
17	axial	−1	0	0	0
18		+1	0	0	0
19		0	−1	0	0
20		0	+1	0	0
21		0	0	−1	0
22		0	0	+1	0
23		0	0	0	−1
24		0	0	0	+1
25	centre	0	0	0	0
26		0	0	0	0
27		0	0	0	0
28		0	0	0	0

References

CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS.